Broadband Optical Reflectance Spectroscopy for multilayer Thin-Film, DBR, and Optical Cavity Analysis

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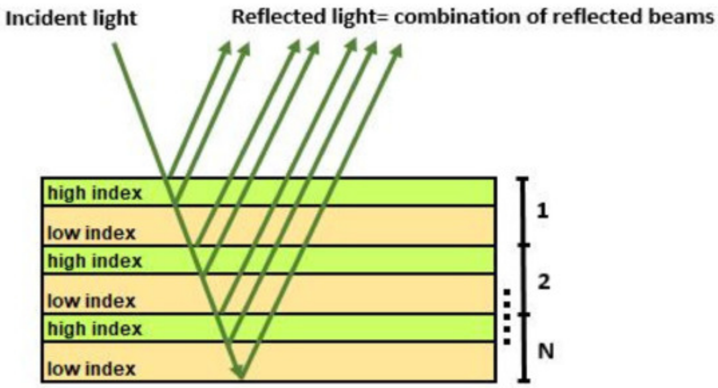
INTRODUCTION

Broadband Optical Reflectance Spectroscopy (BORS) is essential for the non-destructive, nanometer-scale characterization of thin films and multi-layered heterostructures. However, commercial metrology platforms like Filmetrics present steep capital costs and operate as closed-source systems, lacking the hardware-software integration flexibility required for custom research. To overcome these barriers, this project introduces a cost-effective, custom-built BORS system and a dedicated software application developed specifically to support thin-film and optical cavity characterization workflows at the Stern Lab at Northwestern University.

DBR AND OPTICAL CAVITY

A primary application of this custom system is the precise characterization of Distributed Bragg Reflectors (DBRs) and optical cavities. For DBRs—which rely on alternating high- and low-refractive-index dielectric layers to achieve near-perfect reflectivity—direct reflectance measurement is essential to experimentally validate the photonic bandgap, identify the center wavelength, and map the stopband bandwidth. When these DBRs are integrated to form a Fabry-Pérot optical cavity, reflectance spectroscopy becomes the vital tool used to observe cavity resonance modes. Measuring the exact spectral positioning and quality factor of the cavity dip is crucial, as these parameters govern the light-matter interaction enhancement necessary for exploring strong coupling regimes and confined optical modes.



Because the optical response of both DBRs and optical cavities depends strictly on the phase accumulation within each constituent layer, precise thickness fitting is paramount. Variations of thickness in dielectric thin films, such as Silicon Dioxide (SiO2) or Silicon Nitride (Si3N4), can cause significant spectral shifts in cavity resonance or degrade DBR reflectivity. By employing robust mathematical optimization frameworks, such as the Transfer Matrix Method (TMM) coupled with non-linear least-squares fitting algorithms, the software extracts exact physical thicknesses from experimental reflectance data. Ultimately, this project bridges the gap between cost-constrained hardware engineering and high-precision scientific analysis, delivering a versatile metrology software application that advances the local design, optimization, and validation of custom optical cavities.

TRANSFER MATRIX METHOD

To model the electromagnetic response of multi-layered structures, such as Distributed Bragg Reflectors (DBRs) and optical cavities, this project implements the Transfer Matrix Method (TMM). At any planar interface between distinct materials, Maxwell's equations dictate that the tangential components of the electric and magnetic fields must remain continuous. Furthermore, as light propagates through a bulk thin-film layer, it accumulates a wavelength-dependent phase shift determined by the material's refractive index and physical thickness. TMM formalizes this behavior by representing the entire thin-film stack as a total system matrix, M. This characteristic 2 times 2 matrix is constructed by sequentially multiplying two fundamental components: interface transmission matrices(D), which handle boundary conditions, and layer propagation matrices(P), which account for phase accumulation.

$$D_{ij} = \frac{1}{t_{ij}} \begin{pmatrix} 1 & r_{ij} \\ r_{ij} & 1 \end{pmatrix}$$

$$P_j = \begin{pmatrix} e^{-i\delta_j} & 0 \\ 0 & e^{i\delta_j} \end{pmatrix}$$

Because the system consists of multiple stacked thin films, the infinite internal reflections and refractions are highly complex to calculate manually. TMM resolves this by sequentially multiplying the characteristic matrices of each constituent layer to compute a single, global system matrix:

$$M = D_{01} P_1 D_{12} P_2 \dots P_m D_{m,m+1}$$

$$M_{\text{total}} = M_1 \times M_2 \times \cdots \times M_N = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}$$

Where the matrix M can be simplified as:

$$M_j = \begin{pmatrix} \cos\delta_j & \frac{i}{n_j} \sin\delta_j \\ i n_j \sin\delta_j & \cos\delta_j \end{pmatrix}$$

By boundary-coupling this total matrix to the incident medium and the underlying substrate, the complex reflection coefficient can be analytically solved.

HARDWARE SETUP

The BORS setup uses a fiber-coupled hardware architecture optimized for stable spectral acquisition. Illumination is provided by a Thorlabs SLS201L stabilized light source equipped with the Thorlabs SLS201C collimation package, which feeds a broad, uniform beam into the illumination leg of a Thorlabs RP22 reflection probe.

To ensure exact normal incidence and a consistent working distance relative to the target thin films, the probe is secured vertically using a Thorlabs RPS fiber probe stand. Broadband light is directed onto the sample, and the back-reflected signal is collected through the same probe assembly and routed to a Thorlabs CCS175 compact spectrometer. The spectrometer captures and digitizes the optical signal, delivering the raw intensity spectrum directly to the custom software application for processing.

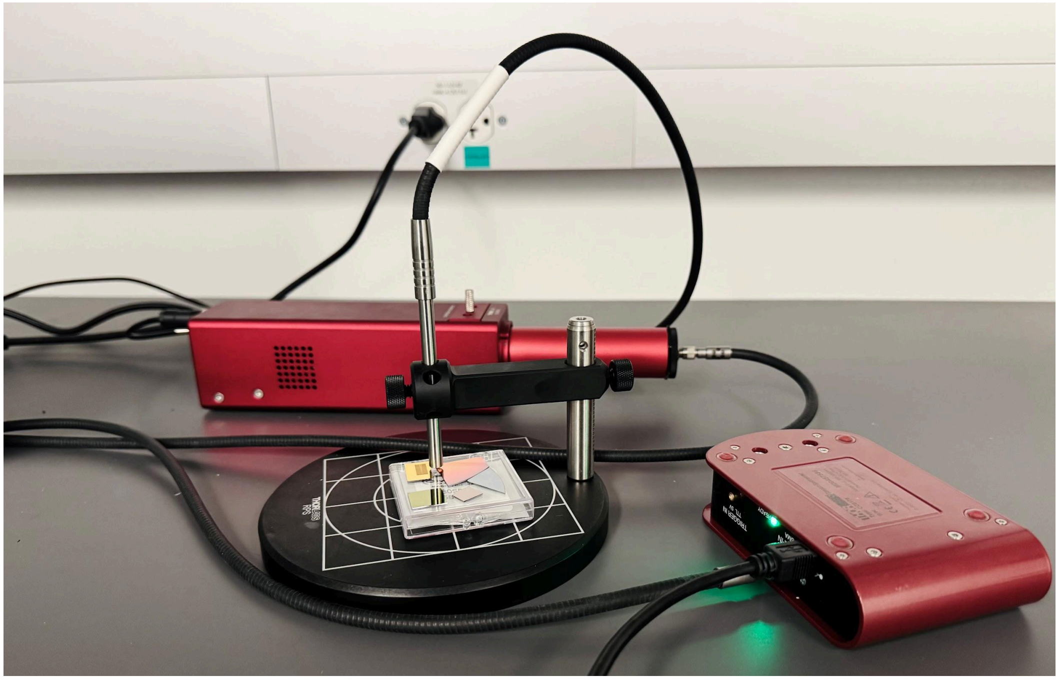


Figure 1. Experimental BORS hardware measurement setup, showing the integration of the Thorlabs SLS201L light source, RP22 reflection probe mounted on the RPS stand, and the CCS175 spectrometer.

EXPERIMENTAL CALIBRATION AND REFLECTANCE NORMALIZATION

To convert the raw optical intensity spectra captured by the spectrometer into absolute reflectance graphs, the software implements a referencing and normalization protocol using a standard silicon (Si) substrate. Because raw spectrometer data is inherently altered by the spectral profile of the SLS201L light source, fiber probe attenuation, and detector quantum efficiency, a correction factor must be established. The software first records the raw intensity spectrum of the unknown sample, a dark background spectrum to account for detector thermal noise, and a reference intensity spectrum taken from a bare silicon wafer. Using the known, theoretically modeled absolute reflectance of silicon (Si) derived via TMM, the wavelength-dependent calibration correction factor is applied to isolate the sample's true response. The final absolute reflectance is computed at each wavelength using the following normalization equation:

$$R = \frac{I_{\text{sample}} - I_{\text{dark}}}{I_{\text{ref}} - I_{\text{dark}}} \times R_{\text{Si}}$$

By mathematically removing the hardware's systemic transfer function, this normalization routine ensures that the resulting data strictly reflects the physical properties of the target thin-film materials, preparing the experimental curves for accurate thickness fitting.

HARDWARE CONTROL AND LABVIEW SYSTEM INTEGRATION

To bridge hardware instrument control with the backend optimization script, a comprehensive control application was engineered in LabVIEW. The architecture imports the native Thorlabs ThorSpectra VI library to initialize and interface directly with the CCS175 spectrometer. The intuitive Front Panel provides a multi-layer configuration layout supporting up to three distinct thin-film layers alongside a selected substrate (e.g., Silicon, Quartz), pulling the corresponding complex refractive index database paths directly within the application.

During operation, LabVIEW displays a live raw intensity graph, allowing the user to visually monitor the detector's signal levels and prevent pixel saturation. To overcome experimental noise and improve the signal-to-noise ratio (SNR) of the raw data, a custom accumulation averaging module was integrated, allowing users to define how many sequential hardware scans to run and average per measurement.

The system guides the operator through a standardized three-step sequential acquisition sequence: recording the dark background noise, the bare silicon reference, and the target sample. Upon completing these three measurements, LabVIEW automatically wakes up and triggers the backend Python script, passing the parsed material arrays and raw intensities. Once the python optimization finishes execution, it feeds the results back to the LabVIEW interface, instantly updating the user display with the final calculated physical layer thicknesses, the statistical Goodness of Fit (GOF), and a synchronized comparison plot of the experimental and fitted reflectance graphs.

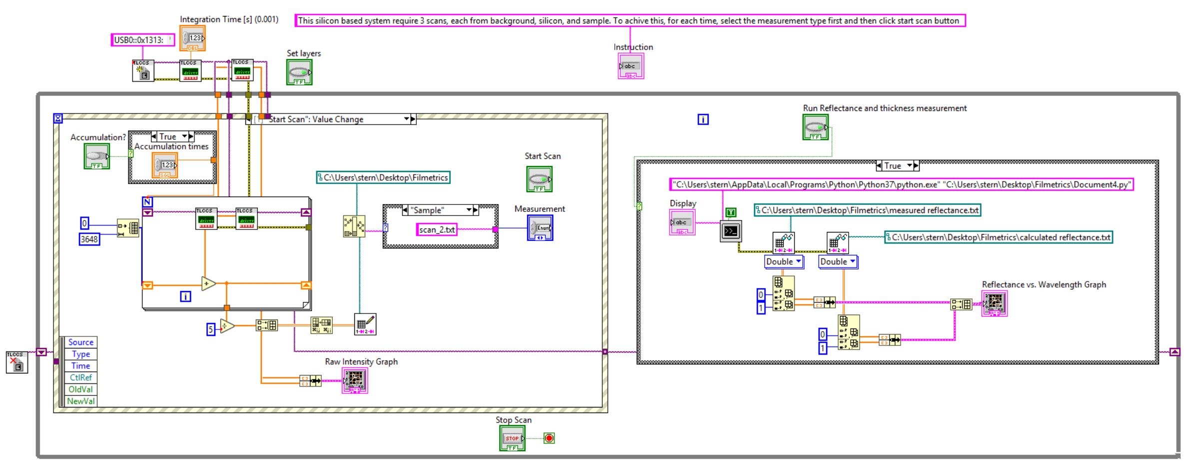


Figure 2. LabVIEW core data acquisition block diagram, showing the integration of the Thorlabs spectrometer drivers, custom accumulation averaging loop, and the backend Python data analysis execution node.

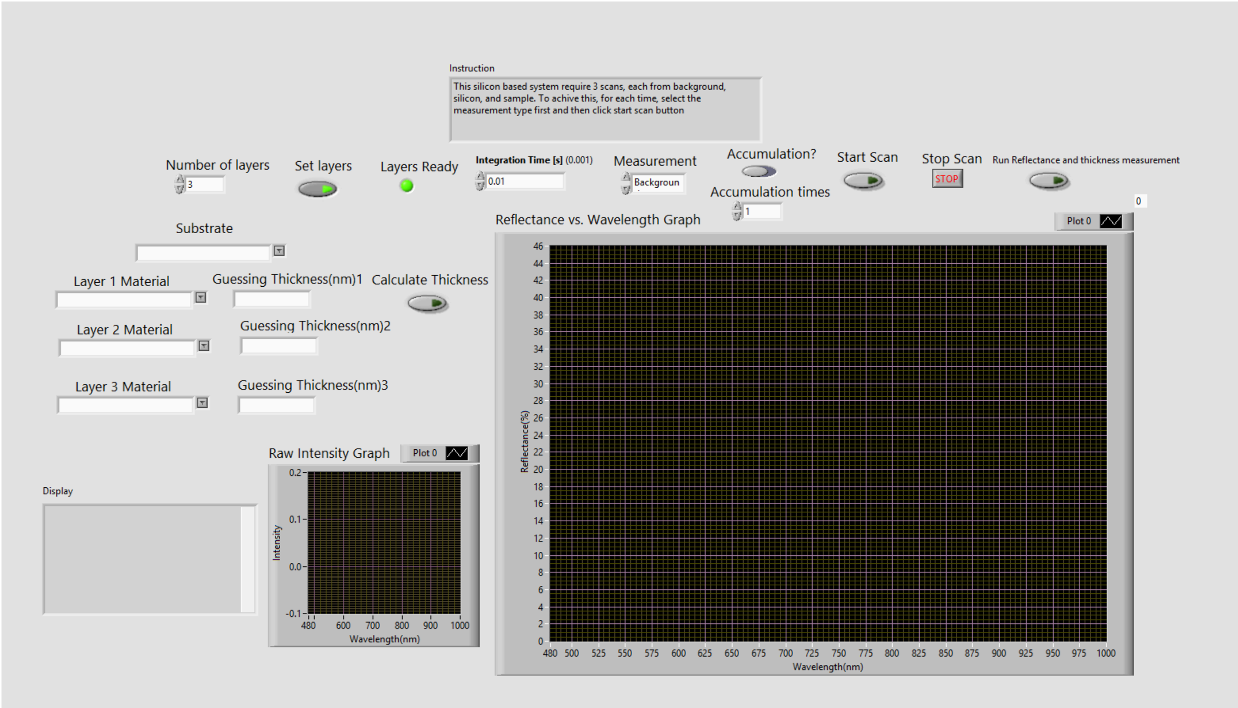


Figure 3. LabVIEW system graphical user interface front panel, showing the material stack inputs, substrate selection, initial thickness guess fields, and the real-time display windows for raw intensity and normalized reflectance curves.

PYTHON SOFTWARE IMPLEMENTATION AND DATA OPTIMIZATION

Once triggered by the LabVIEW front-end, the backend Python script automates a complete computational metrology pipeline divided into three primary functional blocks:

<u>Calibration and Normalization:</u> The software ingests the raw spectrometer intensity data passed from LabVIEW and handles wavelength calibration. It loads the dark background array to eliminate detector thermal noise and computes a wavelength-dependent correction factor by interpolating the raw intensity of the bare silicon wafer against its known theoretical reflectance. This normalizes the raw data, converting it directly into an absolute reflectance spectrum

<u>Transfer Matrix Method (TMM) Forward Model:</u> Utilizing a vectorized wrapping of the tmm library, the code dynamically parses the user-defined thin-film stack and substrate parameters selected in LabVIEW. It pulls the complex refractive index (n) data from the local material database files and computes the polarization-averaged theoretical reflectance across a high-resolution grid

<u>Multi-Start Thickness Fitting:</u> To extract accurate film thicknesses and avoid local minima, the application utilizes a dual-phase optimization engine: it first executes a localized grid-search sweep across an initial parameter space centered on the user’s estimate. The best-performing

winning seed is then passed to a Trust Region Reflective (trf) non-linear least-squares algorithm (scipy.optimize.least_squares), which simultaneously fits the physical layer thicknesses and an intensity scale factor to maximize the statistical Goodness of Fit (GOF) before returning the final parameters to LabVIEW.

MEASUREMENT RANGE

Substrate: Silicon(Si), Quartz(SiO_2), Sapphire
Materials**:** Silicon(Si), Silicon Dioxide(SiO_2), Aluminum Oxide(Al_2O_3), Titanium Dioxide(TiO_2), Tantalum Pentoxide(Ta_2O_5), Silver(Ag), Gold(Au), Poly(methyl methacrylate) 950k (PMMA 950k)
Maximum Layers for Thickness Calculation: 3
Wavelength range: 480nm-1000nm

OPERATING PROCEDURE

1. System Configuration

1a: Input your desired integration time.

1b: If needed, click the Accumulation button and set the number of cycles.

1c: Enter the total number of thin-film layers and select the appropriate substrate from the menu. Click Set Layers and wait for the Layers Ready indicator to illuminate.

1d: Once the indicator is lit, input the material type and an initial thickness estimate for each layer into the generated fields. Click Calculate Thickness to store these values for the fitting algorithm.

2. Data Acquisition

2a: Select the measurement object (Background, Silicon, or Sample) and click Start Scan. Repeat this process for all required objects.

2b: Once all scans are complete, click Run Reflectance & Thickness Measurement.

2c: The application will execute the Python-based fitting routine. The software will display the optimized thickness results, the Goodness of Fit (GOF) value, and a Reflectance vs. Wavelength plot, where the measured data is shown in white and the calculated model in green.

3. Iterative Refinement

To improve the fit, adjust the material parameters or your initial thickness estimates. Click Calculate Thickness again to re-run the fitting engine and compare the updated model against the measured data.

SAMPLE TESTING

a. Pretest simulation on python for reflectance calibration before integrating into LabView using sample Thorlab Mirror PF10-03-P01P.
(Correction factor of 1.04 applied to the measured reflectance for better alignment)

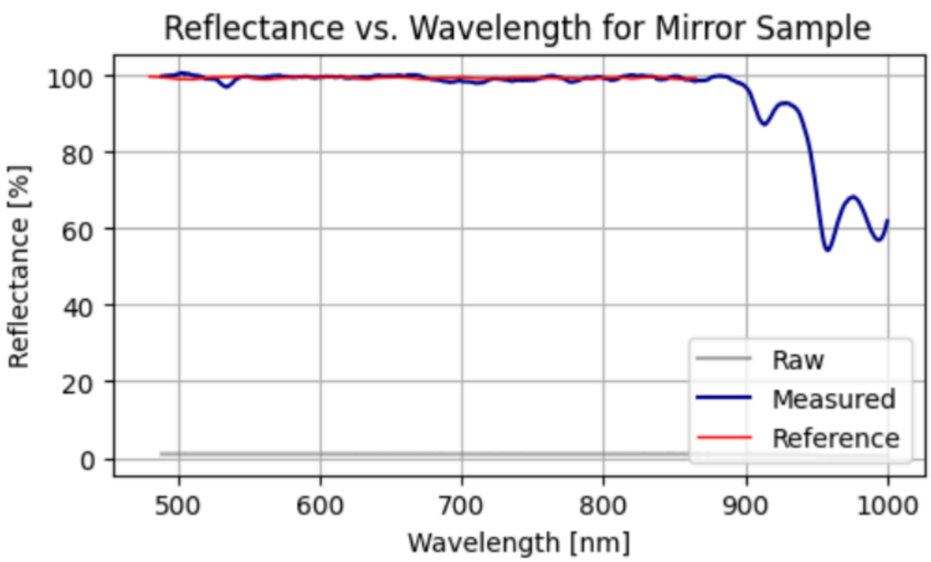


Figure 4. Reflectance vs. wavelength graph for a Thorlabs PF10-03-P01P protected silver mirror, illustrating the overlay of raw measured data (blue) against the theoretical reference curve (red) sourced from the manufacturer's specification data.

b. DBR Reflectance Measurement

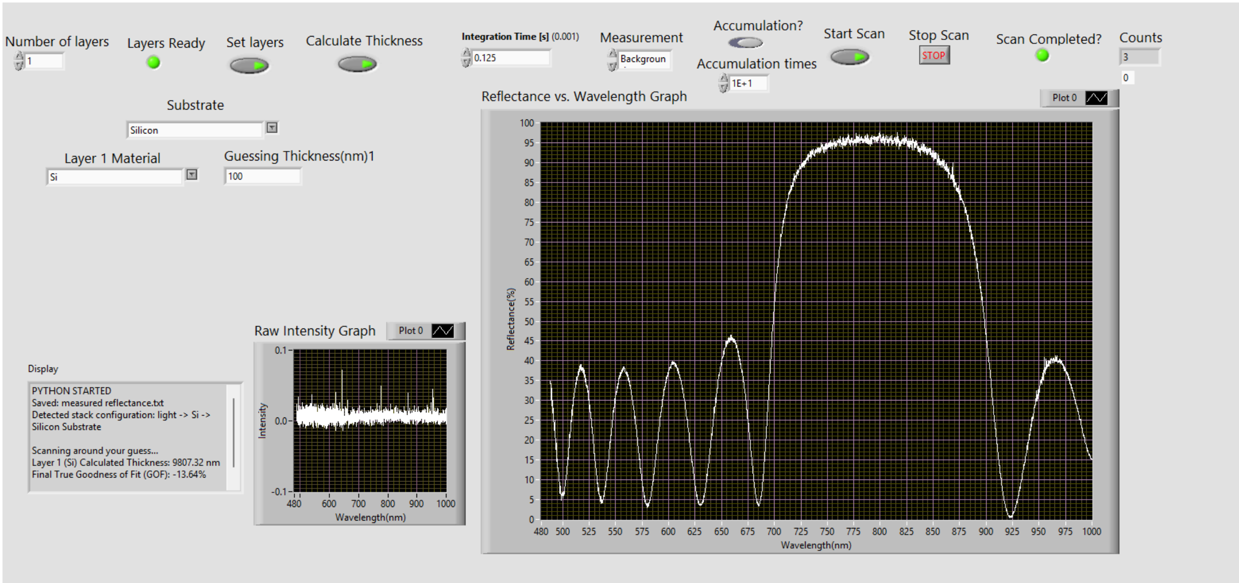


Figure 5. Reflectance vs. wavelength graph for a Distributed Bragg Reflector (DBR), demonstrating the characteristic high-reflectivity stopband with a peak reflectance of approximately 95% at 785 nm.

c. Single-layer SiO2 Reflectance Measurement + Thickness Fitting

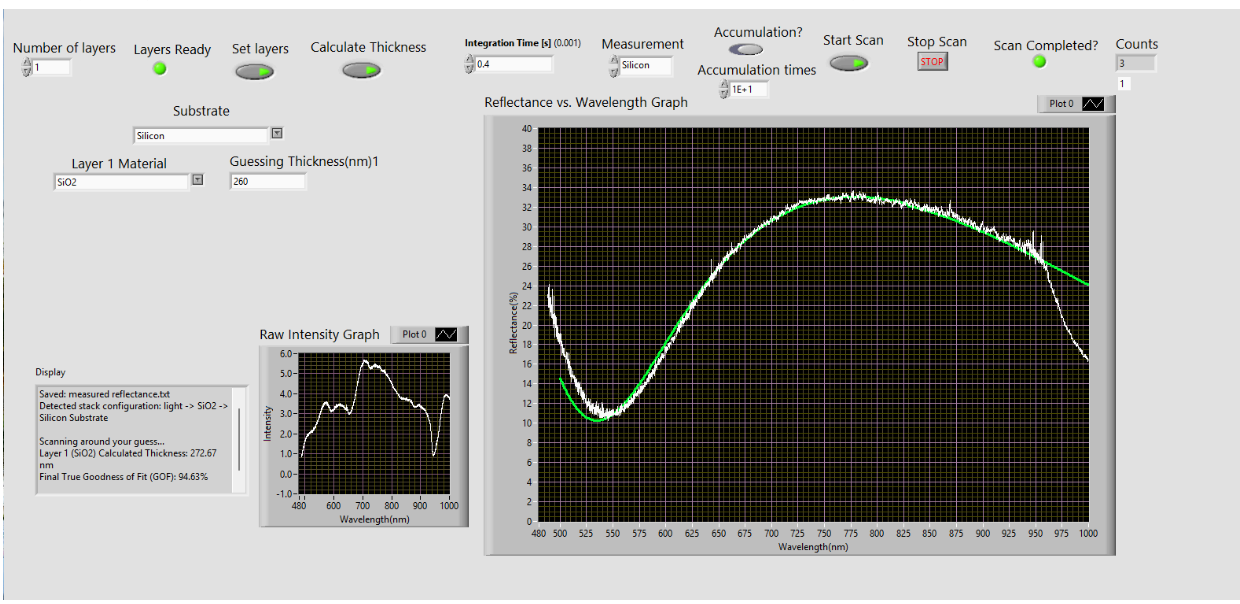


Figure 6. Thickness fitting results for a SiO2 sample on a silicon substrate, showing a precise overlay of the measured reflectance spectrum (white) and the calculated best-fit model (green). The measured reflectance spectrum was validated against the reflectance spectrum simulation. The analysis yields a measured thickness of 272.67 nm, validated against a commercial Filmetrics system, with a Goodness of Fit of 94.63%.

d.Two-layers Si3N4 on top of SiO2 Reflectance Measurement + Thickness Fitting

Figure 7. Reflectance spectrum analysis of a two-layered Si3N4/SiO2 stack on a silicon substrate. The measured reflectance (white) is plotted against the calculated best-fit model (green). The model determines layer thicknesses of 257.10 nm (Si3N4) and 2968.44 nm (SiO2), validated against commercial Filmetrics data. The achieved Goodness of Fit of 99.5% demonstrates the high accuracy and efficiency of the system.

CONCLUSION

In summary, this project demonstrates high accuracy and efficiency in reflectance measurement and thin-film thickness extraction through using data normalization and the Transfer Matrix Method. While current experimental constraints limited validation to two-layer systems, future work will aim to test three-layer stacks (we currently do not have it now) to further confirm the robustness and versatility of the fitting algorithm.

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APPENDIX

Complete Python Code:
https://drive.google.com/file/d/1M9RU5-m2kMFHiL573zd2sD95IOUNVh2Y/view?usp=sharing

Complete Labview Vi:
https://drive.google.com/file/d/1Z71PLSK19aVSw399U1DysvG-JgTMQC_m/view?usp=sharing

Whole Application Package:

https://github.com/Samantha-Q-Cronin/Filmetrics-